Workshop: turn a nonlinear model into an answer

Return to district income. Let I be annual income per person, in thousands of 1998 dollars.

Question

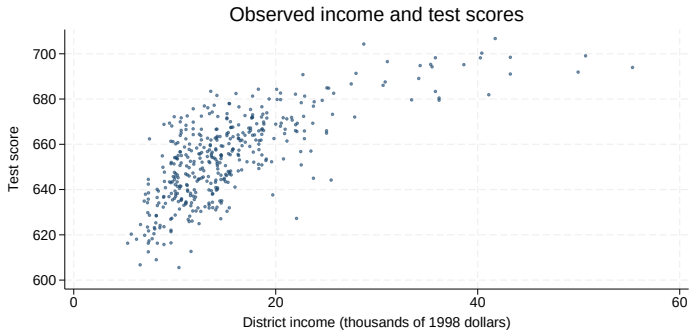
How much does the fitted score change from $I = 10$ to $I = 15$, holding STR and EL fixed?

```
. summarize income
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      income |        420    15.31659     7.22589       5.335     55.328
```

Both incomes are inside the observed range. Does that alone establish a causal comparison?

See where the observations lie

```
twoway scatter score income
```



The scatter is not yet adjusted for STR or EL. A range check does not establish causality or adequate support at every combination of controls.

Write the contrast before looking at the coefficients

For income I in thousands of dollars, consider

$$Score = \alpha_0 + \alpha_1 I + \alpha_2 I^2 + \alpha_3 STR + \alpha_4 EL + u.$$

The α's are population coefficients and u contains other score determinants.

Before estimating, compare income 10 with 15 while holding STR and EL fixed:

Regressor	Before	After	Change
I	10	15	?
I^2	100	225	?
STR and EL	Same	Same	?

Write the fitted difference as a linear combination of $\widehat{\alpha}_1$ and $\widehat{\alpha}_2$. Which terms cancel?

Stata: estimate the controlled quadratic

```
. regress score c.income##c.income str el, vce(robust)
Linear regression                               Number of obs     =        420
------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      income |   2.665477   .2335836    11.41   0.000     2.206323    3.124632
c.income#c.. |  -.0247843   .0042885    -5.78   0.000    -.0332143   -.0163543
         str |  -.2564803   .2952173    -0.87   0.385     -.836788    .3238274
          el |  -.4536707   .0305133   -14.87   0.000    -.5136505   -.3936908
       _cons |   632.6268   5.828567   108.54   0.000     621.1696    644.084
------------------------------------------------------------------------------
```

`c.income##c.income` includes income and income squared.

Both income terms change in a finite comparison. Hold STR and EL fixed.

Report the 10-to-15 contrast with its uncertainty

```
. lincom 5*income + 125*c.income#c.income
 ( 1)  5*income + 125*c.income#c.income = 0
------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   10.22935   .6624466    15.44   0.000     8.927185    11.53152
------------------------------------------------------------------------------
```

$$5(2.665477) + 125(-0.0247843) \approx 10.229 \text{ points.}$$

Holding STR and EL fixed, the fitted score rises by about **10.23 points**, with a 95% interval of about **8.93 to 11.53 points**.

This is an interval for a fitted mean difference, not a prediction interval for a new district.

Your report for a higher-income comparison

The controlled quadratic estimates are $\widehat{\alpha}_1 = 2.665477$ and $\widehat{\alpha}_2 = -0.0247843$. Income is in thousands of dollars.

Your district comparison is now \$30,000 versus \$35,000, holding STR and EL fixed.

1. Write the contrast and calculate the fitted score change.
2. Explain why it differs from 10 to 15, even though both add \$5,000.
3. A colleague writes: "Every \$5,000 increase causes a 10.23-point gain." Rewrite that claim for your comparison.

Have one partner check the arithmetic and the other check the interpretation. Be ready to explain one correction.

Debrief: equal dollar increases need not give equal gains

$$\widehat{\Delta} = 5\widehat{\alpha}_1 + (35^2 - 30^2)\widehat{\alpha}_2 = 5\widehat{\alpha}_1 + 325\widehat{\alpha}_2.$$

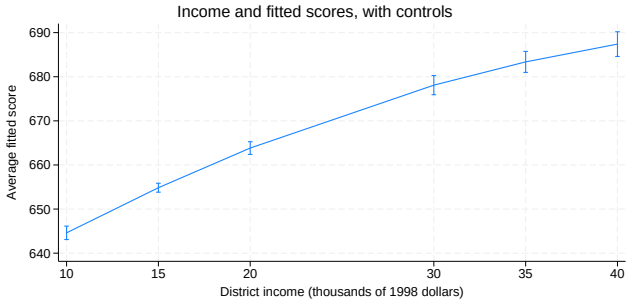
```
. lincom 5*income + 325*c.income#c.income
 ( 1)  5*income + 325*c.income#c.income = 0
------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   5.272501   .3923998    13.44   0.000     4.501162     6.04384
------------------------------------------------------------------------------
```

The fitted gain is about 5.27 points, compared with 10.23 for income 10 to 15. The negative square term reduces the gain more at the higher starting income.

The 95% interval for this fitted difference is about 4.50 to 6.04 score points, with STR and EL held fixed.

See the fitted comparison on the curve

```
margins, at(income=(10 15 20 30 35 40))
marginsplot
```



Pairs: compare the rises from 10 to 15 and from 30 to 35. What do the interval bars show?

The fitted income comparison and its limits

Income change	Fitted score gain	95% interval
\$10,000 to \$15,000	10.23	[8.93, 11.53]
\$30,000 to \$35,000	5.27	[4.50, 6.04]

Both comparisons hold STR and EL fixed. The quadratic permits the gain to depend on the starting income.

Suppose parental involvement affects scores and remains related to income after these controls. Would the fitted curve reveal the causal effect of income?

An endogenous regressor

Let Y be the outcome, X the regressor of interest, and u other outcome determinants in

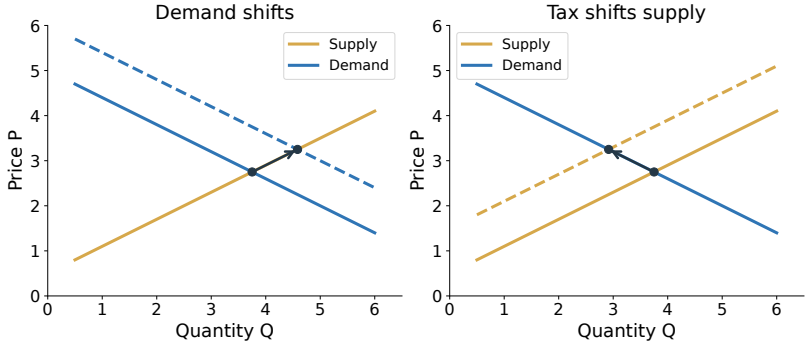
$$Y = \beta_0 + \beta_1 X + u.$$

We call X **endogenous** when $\text{Cov}(X, u) \neq 0$.

Omitted confounders can create this correlation. Simultaneous determination can create it too: market demand shocks change both sales and price.

Adding a square or interaction changes the fitted shape. The correlation with u can remain.

Price variation from demand and supply



A demand shift raises both price and quantity. A supply shift can trace the demand curve when demand otherwise stays fixed.

Which movement would help estimate the response of demand to price?

The idea of an instrumental variable

An **instrument** Z supplies a source of variation in X that can identify its effect on Y.

- **Relevance:** Z changes or predicts X.
- **Exogeneity and exclusion:** Z is unrelated to other outcome determinants and affects Y only through the modeled X channel.

For cigarette demand, a tax is a candidate instrument for price. Anti-smoking attitudes or accompanying restrictions can invalidate that argument.

Next class: evaluate this proposal and estimate the demand elasticity using IV.